Fundamentals 16

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1a. Define $f : \mathbb{R} \to \mathbb{R}$ as

$$f(x) = \begin{cases} 1 & x \geq 0, \\ 0 & x < 0. \end{cases}$$

Regardless of the choice of $K \subseteq \mathbb{R}$, $f(K) \subseteq \{0, 1\}$, which is closed and bounded. So f is a compact map, but not continuous.

2. Since $x, y \in [-1, 1]$, the maximum value of $|x + y|$ is 2. Fix ε and let $\delta < \varepsilon/2$. If $|x - y| < \delta$, then

$$|x^2 - y^2| = |x - y||x + y| < \delta|x + y| \leq 2\delta < \varepsilon,$$

so f is uniformly continuous.

To show that $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$ is not uniformly continuous, we need to show that there exists $\varepsilon > 0$ such that for all $\delta > 0$ there exist $x, y \in \mathbb{R}$ such that if $|x - y| < \delta$ then $|x^2 - y^2| \geq \varepsilon$.

Fix ε and δ, and choose x, y such that $|x + y| \geq \varepsilon/\delta$ and $|x - y| < \delta$. Then we have that

$$|x^2 - y^2| = |x - y||x + y| \geq |x - y|\frac{\varepsilon}{\delta} > \varepsilon,$$

so f is not uniformly continuous.

3a. Fix $\varepsilon > 0$ and let $\delta = \varepsilon a^2$. If $x, y \in [a, \infty]$ then the maximum value of $|xy|$ is a^2. If $|x - y| < \delta$, then

$$\left|\frac{1}{x} - \frac{1}{y}\right| = \left|\frac{y - x}{xy}\right| < \frac{\delta}{|xy|} \leq \frac{\delta}{a^2} = \varepsilon,$$

so f is continuous.

3b. We need to show that there exists $\varepsilon > 0$ such that for all $\delta > 0$ there exist $x, y \in (0, \infty)$ such that if $|x - y| < \delta$ then $\left|\frac{1}{x} - \frac{1}{y}\right| \geq \varepsilon$.

Fix $\varepsilon > 0$ and $\delta > 0$ and choose x, y such that $|xy| < \delta/\varepsilon$ and $|x - y| < \delta$. Then

$$\left|\frac{1}{x} - \frac{1}{y}\right| = \left|\frac{y-x}{xy}\right| \geq |y-x|\frac{\varepsilon}{\delta} > \varepsilon,$$

so f is not uniformly continuous.

4. Consider the restriction of f to $[0,1]$. Since $[0,1]$ is compact and f is continuous, $f : [0,1] \to \mathbb{R}$ is uniformly continuous by Proposition 16.9. Now, we consider f restricted to $(1,\infty)$. We see that, if $x, y \in (1,\infty)$ then $|\sqrt{x}+\sqrt{y}| > 2$. Fix $\varepsilon > 0$ and choose $\delta = 2\varepsilon$. If $|x - y| < \delta$ then

$$|\sqrt{x} - \sqrt{y}| = \frac{|\sqrt{x}+\sqrt{y}||\sqrt{x}-\sqrt{y}|}{|\sqrt{x}+\sqrt{y}|} = \frac{|x-y|}{|\sqrt{x}+\sqrt{y}|} < \frac{\delta}{2} = \varepsilon,$$

so f is uniformly continuous in $(1,\infty)$ as well.

5. Since f is Lipschitz, there exists K such that

$$\rho(f(x), f(y)) \leq K d(x,y)$$

for all $x, y \in X$. Now fix ε and let $\delta = \varepsilon/K$. Then if $d(x,y) < \delta$,

$$\rho(f(x), f(y)) \leq K d(x,y) < K\delta = K\frac{\varepsilon}{K} = \varepsilon,$$

so f is uniformly continuous.

An example of a function that is uniformly continuous but not Lipschitz is $f : [0,\infty) \to \mathbb{R}$ defined by $f(x) = \sqrt{x}$. We know it is uniformly continuous, but it is not Lipschitz because the rate of change near 0 is unbounded.

6. Since f is nonconstant, $f([a,b])$ has at least 2 elements. For contradiction, assume $f([a,b]) = \{u, v\}$ with $u < v$. Then let

$$A = \{x \in [a,b] : f(x) = u\},$$

$$B = \{x \in [a,b] : f(x) = v\}.$$

Take two points $x_0 \in A$ and $y_0 \in B$. WLOG, $x_0 < y_0$. Define

$$c = \inf\{x \in [x_0, y_0] : f(x) = v\}.$$

The set is nonempty because $y_0 \in [x_0, y_0]$ and $f(y_0) = v$, and $c \in [x_0, y_0]$ because it is closed.

Consider the case where $f(c) = u$. Let $\varepsilon = \frac{v-u}{2}$. Since f is continuous, there exists $\delta > 0$ such that

$$|x - c| < \delta \implies |f(x) - f(c)| = |f(x) - u| < \varepsilon = \frac{v-u}{2}.$$

But if $|x - c| < \delta$ and $x > c$, then by the definition of c, $f(x) = v$ and $|v - u| = |f(x) - f(c)| > \frac{v-u}{2} = \varepsilon$, which is a contradiction.

By a similar argument, if $f(c) = v$, then there exists $x < c$ arbitrarily close to c where $f(x) = u$, contradicting continuity of f. Therefore, $f([a, b]) \neq \{u, v\}$, so it contains more than two values.

7. Define $f : \mathbb{Q} \to \mathbb{Q}$ as

$$f(x) = \begin{cases} 0 & \text{if } x < \sqrt{2}, \\ 1 & \text{if } x > \sqrt{2}. \end{cases}$$

To show that this is continuous, consider any $x_0 \in \mathbb{Q}$. Given any ε we can let $\delta = |x_0 - \sqrt{2}|$, so that if $|x - x_0| < \delta$ then $f(x) = f(x_0)$.

But this doesn't satisfy the intermediate value property because, if we let $y = \frac{1}{2}$ there is no $x \in \mathbb{Q}$ such that $f(x) = y$.

8a. Because $\sin(x)$ and $\frac{1}{x}$ are continuous whenever $x \neq 0$, $\sin\left(\frac{1}{x}\right)$ is continuous too. So if $0 \notin [a, b]$ then by IVT, the intermediate value property is satisfied.

Assume that $0 \in [a, b]$. If we need a value $y \in [-1, 1]$, let $t_0 = \arcsin(y)$. Then for infinitely many integers n, $t_n = t_0 + 2\pi n$ also satisfies $\sin(t_n) = y$. Now set $x_n = 1/t_n$, so that $x_n \to 0$ as $n \to \infty$. So, arbitrarily close to 0, f takes the value y. Therefore, since $0 \in [a, b]$, given any y, we can find some $x \in [a, b]$ such that $f(x) = y$.

8b. Since $\sin(1/x)$ is noncontinuous but satisfies the intermediate value property for all intervals $[a, b]$, the converse of IVT is false.

9. Consider the last point at which the success rate is $< \frac{4}{5}$, and say there were a successes and b total trials at that point. Then $\frac{a}{b} < \frac{4}{5}$ and $\frac{a+1}{b+1} \geq \frac{4}{5}$. Simplifying this inequality, we get

$$\frac{a+1}{b+1} \geq \frac{4}{5}, \quad 5(a+1) \geq 4(b+1), \quad 4b - 1 \leq 5a.$$

Since $\frac{a}{b} < \frac{4}{5}$, we also have that $5a < 4b$, so

$$4b - 1 \leq 5a < 4b, \quad -1 \leq 5a - 4b < 0.$$

Since a and b are integers, $5a - 4b$ is an integer, so $5a - 4b = -1$. So we have that

$$5a - 4b = -1, \quad 5(a+1) - 4(b+1) - 5 + 4 = -1,$$
$$5(a+1) = 4(b+1), \quad \frac{a+1}{b+1} = \frac{4}{5}.$$

So the success rate is exactly 80% at this point.